

Ticklish Little Guy

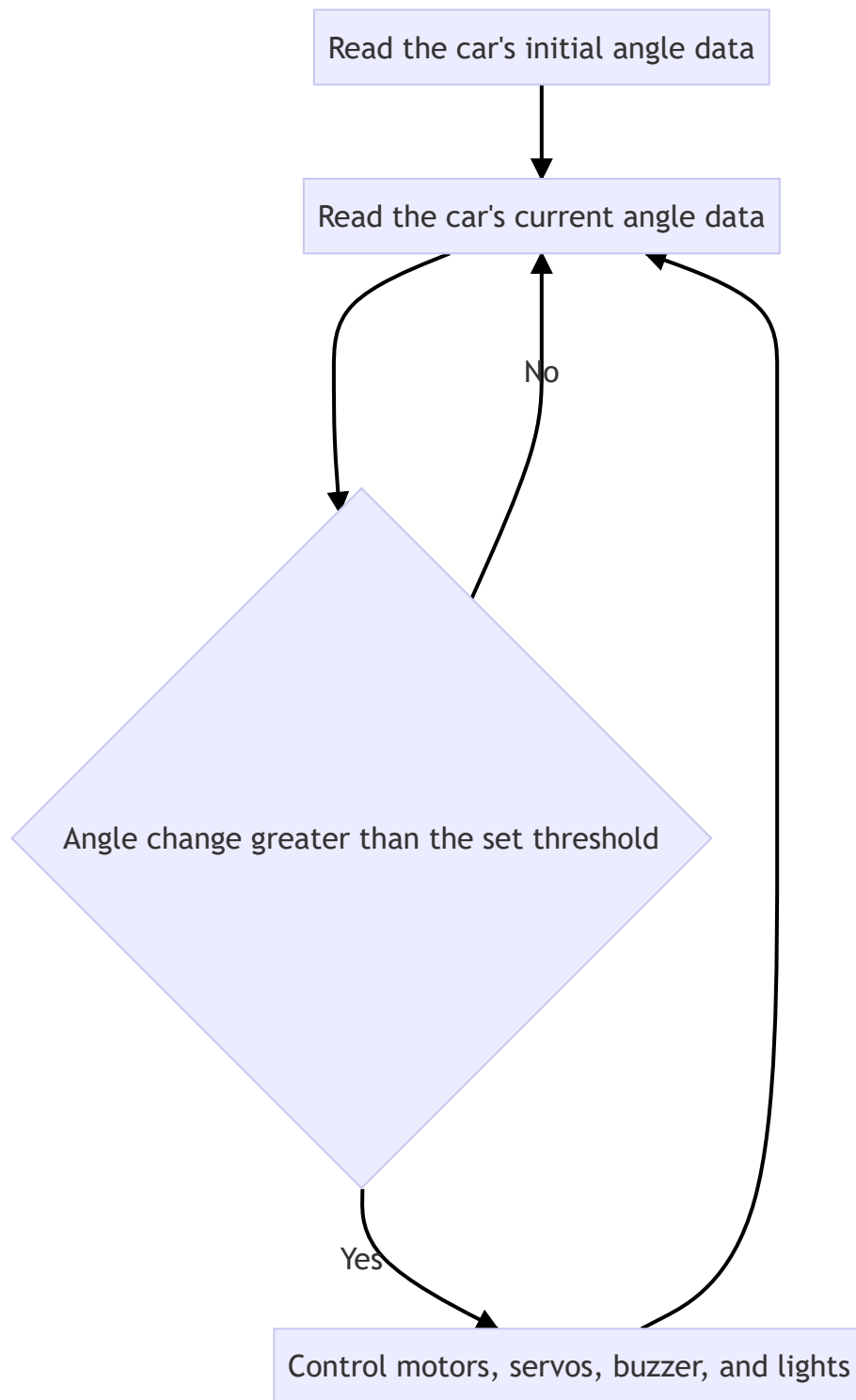
Ticklish Little Guy

1. Experiment Purpose
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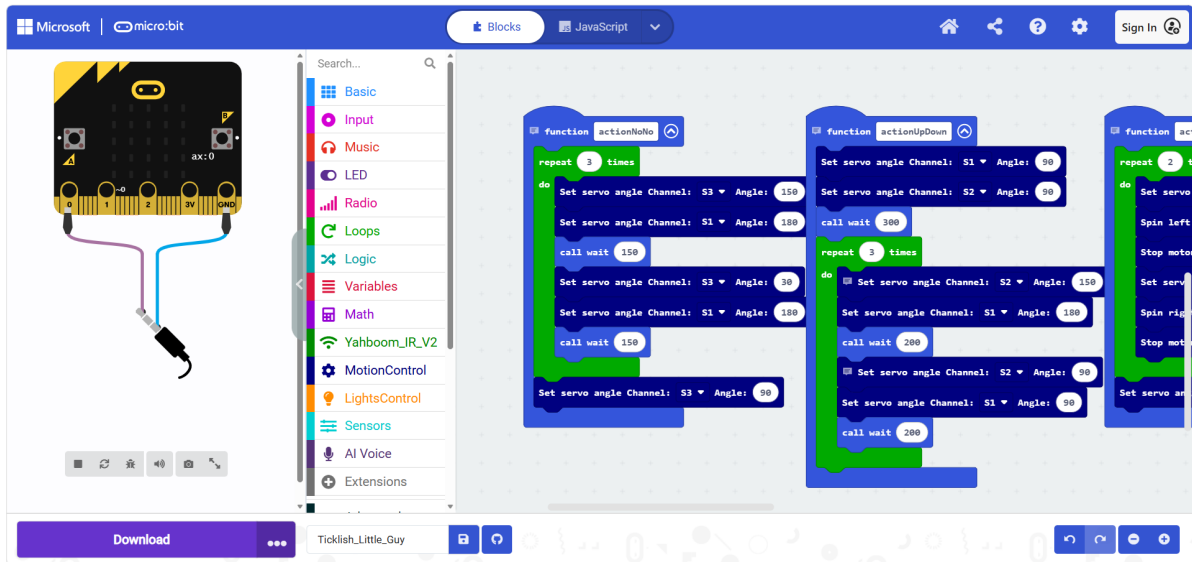
1. Experiment Purpose

The accelerometer is a sensor that measures movement. The accelerometer in the micro:bit can detect the car's movement direction at any time. In this section, we will learn to use the accelerometer sensor to detect the car's movement and achieve the "ticklish" effect. When the car is detected to be tilted (simulating being tickled), the car will perform a series of interesting actions, including moving up and down, looking left and right, testing with the gripper, shaking the head to refuse, and swaying the head in small steps.

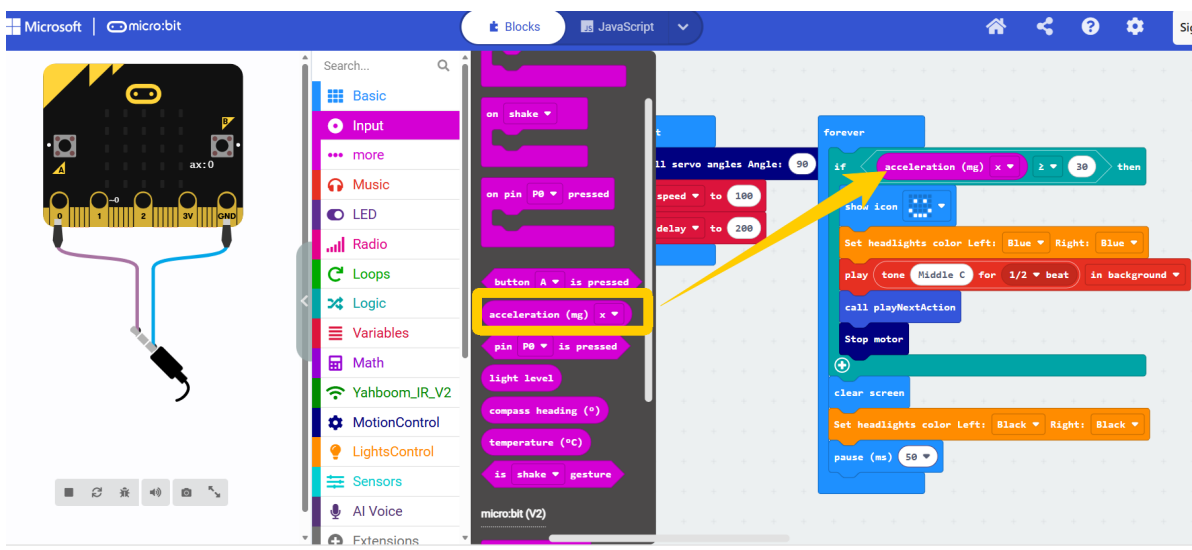
2. Implementation Principle



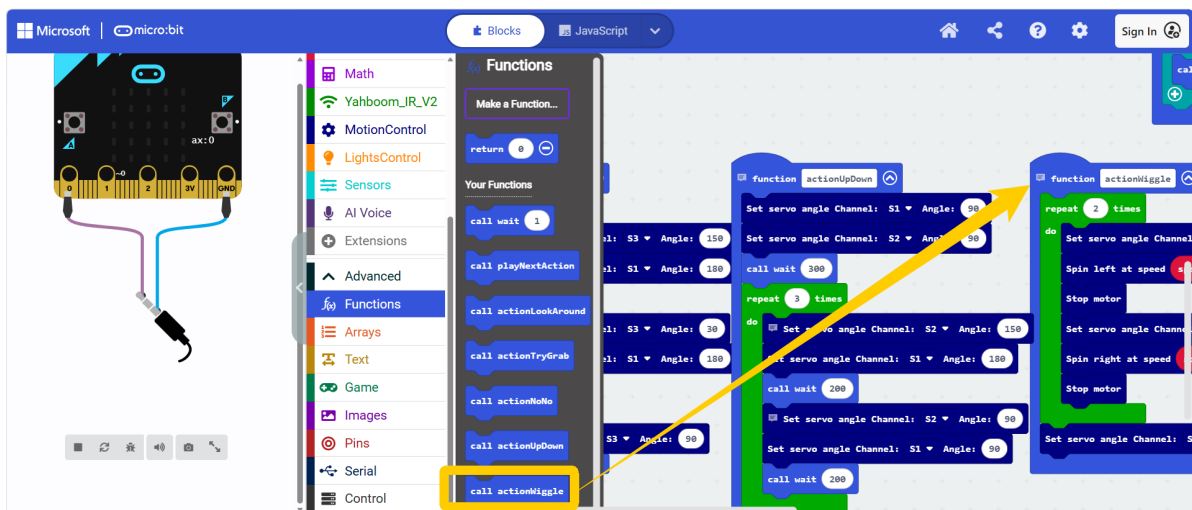
3. Experiment Source Code



Reading the accelerometer value is located in the **Input** category.



The functions are located in the **Functions** subcategory under the **Advanced** category.



The positions of other blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.

2. Flash `8.Ticklish_Little_Guy.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

When the car is detected to be shaken left and right, it will show a happy expression, turn on the blue headlights, play a prompt tone, and then perform five interesting actions in sequence: moving up and down, looking left and right, testing with the gripper, shaking the head to refuse, and swaying the head in small steps. Each time it is gently tilted, the action switches to the next one.

